
FNH 200 942

Lesson 13 - Food, Nutrition, and Health

Student Experience of Instructions, please, please!

Content first, then review 'assessment' questions

2026 Summer

Do you have a couple bottles of these at home? What do you take?



Which country led the way in the regulation of functional foods?

- A. Canada
 - B. China
 - C. Japan
 - D. Korea
 - E. The United States
-

Which country led the way in the regulation of functional foods?

Japan

Food for Specific Health Use (FOSHU)

Japan - FOSHU

Canada

Natural and Non-Prescription Health Products

This is an example of...



- A. Functional Foods
- B. Natural Health Product
- C. Nutraceuticals
- D. Nutritionals
- E. None of the above

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What are natural health products?

- (i) Vitamins, minerals, amino acids, and fatty acids
- (ii) Herbal remedies
- (iii) Homeopathic medicines
- (iv) Traditional medicines from different culture
- (v) Probiotics

- A. (i)
 - B. (ii), (iii)
 - C. (iii), (iv)
 - D. (ii), (iii), and (iv)
 - E. All
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Natural Health Products

Under the *Natural Health Products Regulations*, which came into effect on January 1, 2004, natural health products (NHPs) are defined as:

- Probiotics
- Herbal remedies
- Vitamins and minerals
- Homeopathic medicines
- Traditional medicines such as traditional Chinese medicines
- Other products like amino acids and essential fatty acids

NHPs must be safe to use as over-the-counter products and do not need a prescription to be sold. Products needing a prescription are regulated as drugs under the *Food and Drug Regulations*.

in Canada

<https://www.canada.ca/en/health-canada/services/drugs-health-products/natural-non-prescription.html>

How do we know this product has been authorized for sale in Canada?

- A. It has 'Canadian' on the label
- B. It has an approved health claim



- C. The manufacturer is reliable.
 - D. The manufacturer has been in operation since 1922.
 - E. It has a NPN (or DIN-HM) number
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Licensed Natural Health Product Database

- product name
- product licence holder
- Natural Product Number (NPN) or Homeopathic Medicine Number (DIN-HM)
- product's medicinal ingredients
- product's non-medicinal ingredients
- product's dosage form
- product's recommended use or purpose (i.e. its health claim or indication)
- risk information associated with the product's use (i.e. cautions, warnings, contra-indications and known adverse reactions)

<https://www.canada.ca/en/health-canada/services/drugs-health-products/natural-non-prescription/applications-submissions/product-licensing/licensed-natural-health-products-database.html>

Search the database

Search criteria

Current status

Simple search

Simple Search allows only one search field criterion be specified, where Advanced Search allows multiple search field criteria however, once you have selected a field, you must enter a criteria for that field. The condition drop down list in Advanced Search allows relationship between each search field criterion.

NPN/DIN-HM: Natural Product Number (NPN) or Homeopathic Medicine Number (DIN-HM)

Search field	Operator	Criterion
--Select Field--	Contains	

To add date criteria to the search field criteria entered, select a condition



NPN 80031182

<https://health-products.canada.ca/lnhpd-bdpsnh/index-eng.jsp>

Examples of functional foods:

(i)

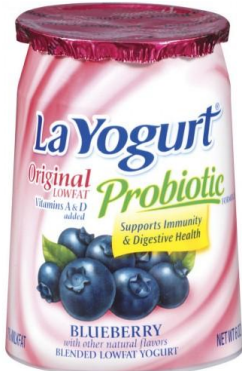


(ii)



(iii)

(iv)



(v)



- A. (i)
- B. (i) and (ii)
- C. (i), (ii), (iii)
- D. (i), (ii), (iii), and (iv)
- E. All of them

Examples of Canadian Functional Foods



Examples of Canadian Natural Health Products

Natural Health Products

- Pacific Agri-Food Research Centre - extraction of anthocyanins from berries and red wine.
 - Nutra Canada - fruit and vegetable extracts.
 - Greenfield Naturals - protein and fibre from by-products of corn-ethanol.
 - Nanton Nutraceuticals - Rhoziva from *Rhodiola rosea*.
 - Island Abbey Foods - Honibe honey lozenges with menthol and eucalyptus.
 - Natural Factors Nutritional Products - PGX for appetite control.
 - Valeant Pharmaceuticals (formerly C.V. Technologies) - Cold-FX from North American ginseng.
-

Ways to develop (make/create) functional foods.

- By adding vitamins and/or minerals
 - By adding bioactive ingredients
 - Produce foods with increased levels of bioactive ingredients through breeding and/or genetic modifications
 - Produce food with increased levels of bioactive ingredients through processing and/or livestock feeding
-

What are some known benefits of probiotics?

- Reduce gastrointestinal discomfort and disorder
- Control irritable bowel syndrome
- Prevent development of colon cancer

Many more potential benefits under investigation

Is *Bifidus regularis* an approved health promoting probiotics added to yoghurt?



- A. Yes
- B. No

What is the minimum levels of probiotics needed for products to make probiotic claims?

1.0 x 10⁹ units of approved probiotics must be present in each serving of product

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To have enough for health benefit:

- 1.0 x 10⁹ units per mL
 - 100 mL, twice a week
-

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completion of the fermentation process.

As you will remember from Lesson 9, fermented food products provide many nutritional advantages such as ease of digestibility and improved availability of some nutrients. Scientific studies have suggested that by stimulating the growth of bifidobacteria and some other probiotics gastrointestinal disorders, intestinal discomfort, and flatulence can be reduced. There are even suggestions that other serious health problems, such as diarrhea, irritable bowel syndrome, colitis, and even the development of colon cancer may be prevented by consuming sufficient quantity of foods **containing viable (live) probiotics**. It is believed that the beneficial properties of probiotics against colon cancer are associated with the metabolic conversion, degradation and/or absorption of carcinogenic compounds as well as stimulation of the immune system.

However, these probiotic bacteria (such as *Lactobacillus acidophilus* and *Bifidobacterium bifidus*.) must be present in numbers high enough to have a physiological effect on the consumer. These numbers should be above 10^6 (1 million) viable organisms per ml and at least 100 ml of the product should be consumed twice per week. Survival of the probiotic cultures during distribution, retailing and in the consumer's home is required to maintain efficacy of those food products (probiotic yogurts) or natural health products (capsules of probiotic bacteria).

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FNH 200

Last but not least... a few more notes

- Final exam format
 - Sample short-answer questions with answers
 - Judy's attempt to understand how teamwork works
-

Final Exam on August 21

Starts between 5 am and 12 noon

- Individual stage - Lockdown Browser required
 - All team members should start at about the same time (5 min of each other)
 - 46 MCQs (include those from team projects, check them after Aug 17)
 - <https://wiki.ubc.ca/Course:FNH200/Projects/2026>
 - 2 short-answer questions (won't be graded until later)
 - 100 minutes
 - The grade you will see will be out of 46 (based on the MCQs only)
- Team stage:
 - Complete together
 - 20 to 25 MCQ, no short answer question
 - 40 minutes

The exam will close at 12 noon; the mark you will see is out of 46

Final Exam - Sample Short Essays

Technical Notes:

- Canvas **automatically** save your typing every few seconds.
 - Near the end, if you keep on typing down to the very last second of the exam, Canvas may tell you that your response has NOT been saved.
 - Don't panic!!!!
 - Good news! It is saved, at least once every few seconds.
 - The score you will see is **only out of 46**, based on the MCQs only
-

Questions?

Sample Short-Essay Questions:

You plan to make a potato salad for your grandpa's birthday. When you get home after a long day of final exams, to your surprise, you find out that the potatoes have turned green. You only have an hour left before the party and you don't have the time (and money!) to prepare something else.

- a. What is the matter? Why is there a concern? (2 marks)
- b. You thought about trimming the green portions of the potatoes off when preparing the salad. How will it affect the level of hazard that you are about to expose to your grandpa's health? (2 marks)
- c. What would you do? Explain. (1 mark)

Note on our marking scheme:

- What you will see on the next slide is a marking scheme that we expect; however, I also ask my TAs to adjust it (make it harder or easier) based on the overall performance of the class.
-

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- c. What would you do? Explain. (1 mark)

a. Potatoes turned green contains higher than normal **solanin** (0.5) at **80 to 100 mg per 100 gram** of potato (compare to 4 to 6 mg per 100 gram) (0.5).

Solanine is a **cholinesterase inhibitor** and may affect **nerve function** after an **oral dose of about 100 mg**.

(include at least two, any two, info **those in green** above to earn 1 mark)

b. Because solanine cannot be destroyed by heat (0.5), and that the solanine concentrates in the green portion (0.5) of potato, trimming the green part (0.5) of the potato is an effective way (0.5) to lower the hazard (0.5) exposed to grandpa.

(exact mark distribution may differ)

C. I would rather **not** bring any potato salad to my grandpa's party. As he is getting old with a weak immune system. He likes the salad so much and will properly consumer a lot. (**must DECLARE a position**, yes or no, then explain to earn mark)

A camping supply company is developing a quality pasta in tomato sauce for hikers going on to multiple-day hikes.

What preservation techniques would you use to prepare this new dish? Why? (3 points)

What packaging materials would you use? Why? (2 points)

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Possible answer #1:

Dehydration (freeze dried or vacuum microwave dried) (1). The use of vacuum enables the removal of water at a lower temperature (0.5) (What method and how it works), (add some reasons) preventing quality loss (0.5). Once water is removed, the dehydrated pasta may also be lighter to carry (0.5) for a multiple-day hike. Freeze-dried (or vacuum microwave dried) products are also easier to be rehydrated (0.5) due to the void (or puff) structure created during the dehydration).

The product should be packaged with material that is a strong water barrier such as plastic (1); light weigh for hiker (1).

A camping supply company is developing a quality pasta in tomato sauce for hikers going on to multiple-day hikes.

What preservation techniques would you use to prepare this new dish? Why? (3 points)

What packaging materials would you use? Why? (2 points)

Possible answer #2, not best, only part mark, I do hope that students recognize the need to create something easy to carry on a hiking trip:

Commercial sterilization (1) to eliminate all possible pathogens (0.5) but the high temperature may affect its quality (0.5). Easy to consume in the wilderness (0.5)

Retort pouch pre-sterilized with hydrogen peroxide OR pasta and sauce in pouch then heated (1); for light weight (1) for hiking.

Not acceptable: pasteurization, freezing, irradiation, can, glass

What have you learned?

1. Project
2. Assignment #3
3. Peer Evaluation

Our In-depth Investigations [\[edit | wikitext \]](#)

Team Number ↕	Product
0 (Example)	The Processing and Safety of Swiss Cheeses
1	Commercially Produced Jam vs Homemade
2	Safety regulations between Canada and Un
3	Plant-Based Egg
4	
5	The Processing and Chemical Reactions of
6	Protein Powder
7	Tempeh and the Art of Traditional Fermenta
8	Olestra Regulation
9	Milk:Fat Content Regulations and Labelling
10	
11	
12	Milk and Ultrafiltered Milk
13	
14	

Between now and next week:

Student Experience of Instruction, please!

Team Project

Peer Review of your teammates (mandatory for grades)

Assignment #3 - Your favourite food

Final Exam

I hope this is just the beginning of your food exploration!

My quest on learning how **TEAMWORK** works

Dear students,

I truly believe that we (human beings, social creatures) learn best when we have a chance to exchange ideas, enlightenments, confusions, frustrations, or any learning process with peers. While I 'push' to have team work in FNH 200, I'm fully aware that the short, intense, and online nature of this summer section may not be the most ideal learning environment to have teamwork.

Please help me learn how I can do better to support your learning in team environment by (1) completing an anonymous survey and/or (2) participating in focus group discussion. All the these data will be gathered and viewed by a PhD student in Education after your final grade is submitted. I will only view the data when all identifiers have been removed (again after all grades are submitted).

Once I analyze your responses, I plan to sharing my learning with other instructors who also believe in teamwork.

Thanks for your support in helping me and other instructors improve our teaching.

Regards,

Judy

Last but not least... Thank you!



Please take care of yourselves and family members

I look forward to bumping into you on campus.

Come find me, we can go for latte and have a chocolate croissant (on me!)

Thank you again for choosing to explore foods with me this summer! It's a true pleasure!

Know our foods, their roots, and how we connect...

